

Chapter 2: Image and Video Processing

IMP302m

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2.1 Basic Linear Filtering with Application to Image Enhancement

- Linear system theory is foundational in digital image and video processing.
- Techniques for modifying, improving, or representing digital visual data are rooted in linear systems concepts.
- Utilizations
 - Image/video contrast improvement, denoising, and sharpening.
 - Target matching and feature enhancement.

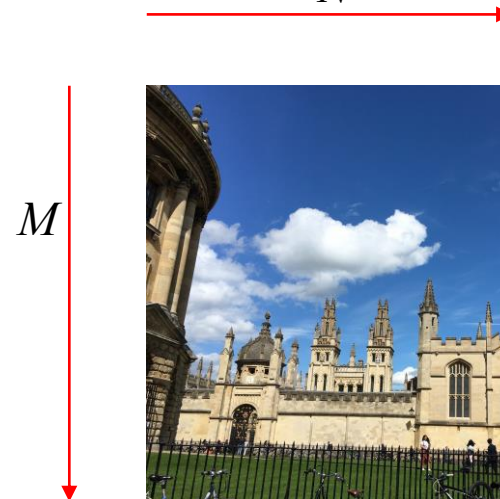
2.1 Basic Linear Filtering with Application to Image Enhancement

- The basic two-dimensional discrete-space signal is the two-dimensional impulse function

$$\delta(m - p, n - q) = \begin{cases} 1 & \text{if } m = p \text{ and } n = q \\ 0 & \text{otherwise} \end{cases}$$

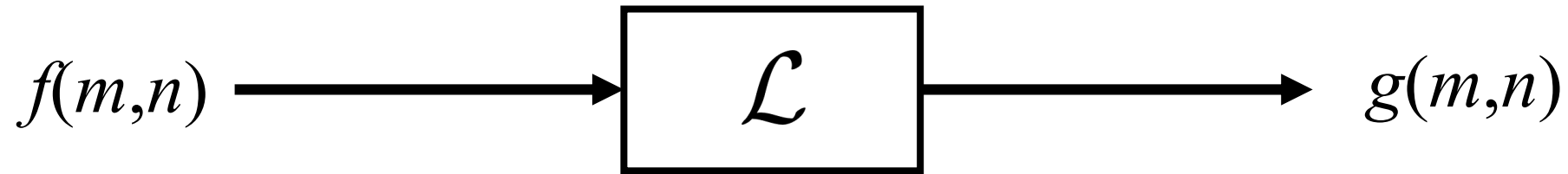
- Any image f can be expressed in terms of the impulse function N

$$\begin{aligned} f(m, n) &= \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(m - p, n - q) \delta(p, q) \\ &= \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) \delta(m - p, n - q) \end{aligned}$$



2.1 Basic Linear Filtering with Application to Image Enhancement

- A two-dimensional system $\mathcal{L}$ is a process of image transformation



$$g(m, n) = \mathcal{L}[f(m, n)]$$

- The system $\mathcal{L}$ is *linear* if and only if for any $f_1(m, n)$ and $f_2(m, n)$ such that $g_1(m, n) = \mathcal{L}[f_1(m, n)]$ and $g_2(m, n) = \mathcal{L}[f_2(m, n)]$ and any two constants a, b , then for every (m, n)

$$a \cdot g_1(m, n) + b \cdot g_2(m, n) = \mathcal{L}[a \cdot f_1(m, n) + b \cdot f_2(m, n)]$$

2.1 Basic Linear Filtering with Application to Image Enhancement

- New image after an image is filtered is

$$g(m, n) = \mathcal{L}[f(m, n)] = \mathcal{L} \left[\sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) \delta(m - p, n - q) \right]$$

It is *linear space-invariant* (LSI), which is invariant to space, then

$$\begin{aligned} g(m, n) = \mathcal{L}[f(m, n)] &= \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) \mathcal{L} [\delta(m - p, n - q)] \\ &= \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) h(m - p, n - q) \\ &= f(m, n) * h(m, n) \end{aligned}$$

2.1 Basic Linear Filtering with Application to Image Enhancement

- Change to frequency domain (Fourier transform)

$$G(U, V) = F(U, V)H(U, V)$$

2.1 Basic Linear Filtering with Application to Image Enhancement

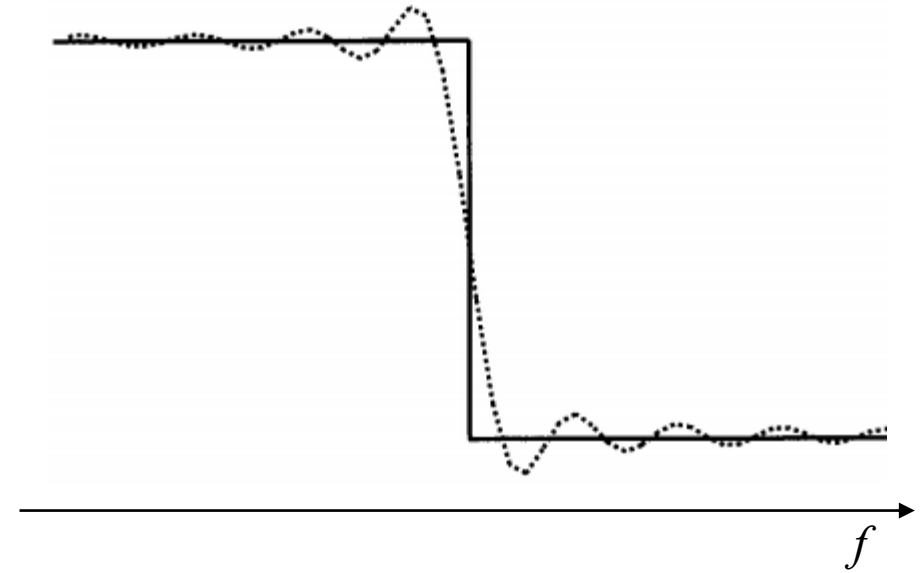
- Ideal Low-Pass Filter: forcing the frequency response to be zero outside of a given radial cutoff frequency Ω_c

$$H(U, V) = \begin{cases} 1 & \text{if } \sqrt{U^2 + V^2} \leq \Omega_c \\ 0 & \text{otherwise} \end{cases}$$

or outside of a rectangle defined by cutoff frequencies along the U- and V-axes

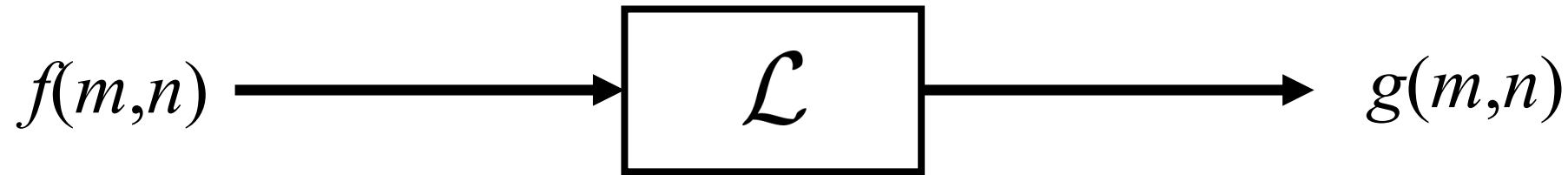
$$h(m, n) = U_c V_c \text{sinc}(2\pi U_c m) \cdot \text{sinc}(2\pi V_c n)$$

$$H(U, V) = \begin{cases} 1 & \text{if } |U| \leq U_c \text{ and } |V| \leq V_c \\ 0 & \text{otherwise} \end{cases}$$



2.1 Basic Linear Filtering with Application to Image Enhancement

- Gaussian Filter



$$h(m, n) = \frac{1}{2\pi\sigma^2} \exp \left[-\frac{(m^2 + n^2)}{2\sigma^2} \right]$$

- Frequency response

$$H(U, V) \approx e^{-2\pi^2\sigma^2(U^2+V^2)} \text{ for } |U|, |V| < \frac{1}{2}$$

2.2 Nonlinear Filtering for Image Analysis and Enhancement

- Effectively preserve edges and fine details in images.
- Less susceptible to noise, maintaining image quality in noisy conditions.
- Noise is inherent in image acquisition due to physical randomness in systems.
 - Common in analog photography, leading to film-grain noise.
 - Both noise and the image signal are captured, affecting the final digital image.

2.2 Nonlinear Filtering for Image Analysis and Enhancement

Median Smoothers

- The observation vector $\mathbf{x}(n) = [x(n - N_L), \dots, x(n), \dots, x(n + N_R)]^T$
 $N = N_L + N_R + 1$, without loss of generality, hence the samples in the observation window are sorted, the median smoother

$$y(n) = \begin{cases} x_{(\frac{N+1}{2})} & \text{if } N \text{ is odd} \\ \frac{x_{(\frac{N}{2})} + x_{(\frac{N}{2}+1)}}{2} & \text{otherwise} \end{cases}$$

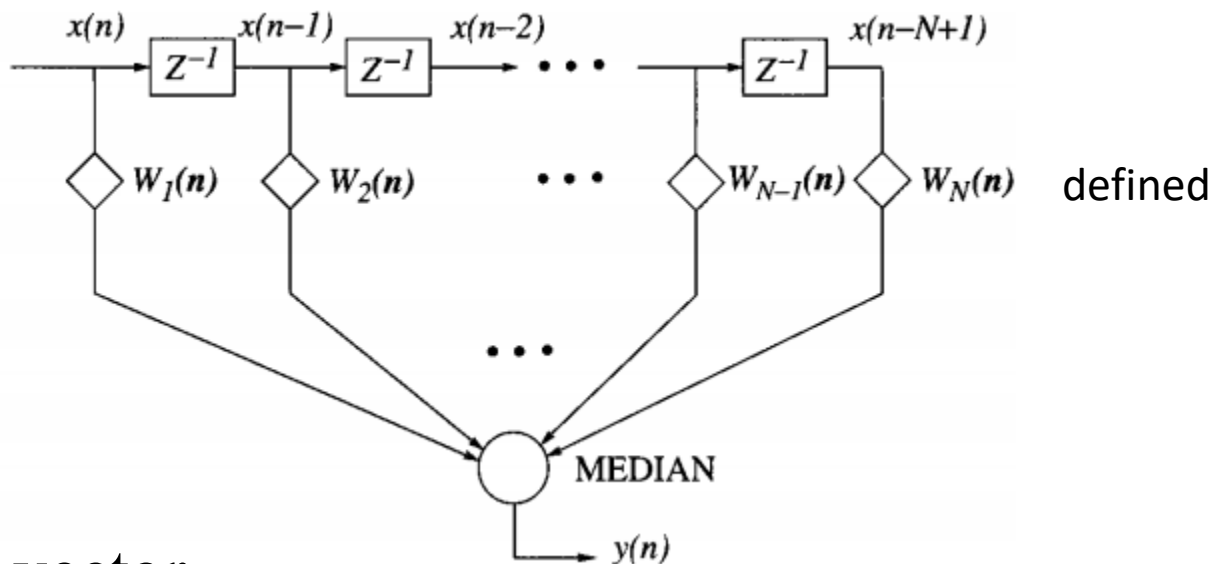
2.2 Nonlinear Filtering for Image Analysis and Enhancement

Weighted Median Smoothers

1. Calculate the threshold

$$W_0 = \frac{1}{2} \sum_{i=1}^N W_i$$

2. Sort the samples in the observation vector
3. Sum the weights corresponding to the sorted samples beginning with the maximum sample and continuing down in order;
4. The output is the sample whose weight causes the sum to become $\geq W_0$



Threshold is 0.35

observation samples	12,	6,	4,	1,	9
corresponding weights	0.1,	0.1,	0.2,	0.2,	0.1
sorted observation samples	1,	4,	6,	9,	12
corresponding weights	0.2,	0.2	0.1,	0.1,	0.1
partial weight sums	0.7,	<u>0.5</u> ,	0.3,	0.2,	0.1

Output is 4

2.2 Nonlinear Filtering for Image Analysis and Enhancement

- Weighted Median Filter

1. Calculate the threshold $T_0 = \frac{1}{2} \sum_{i=1}^N |W_i|$
2. Sort the "signed" observation samples $\text{sgn}(W_i)X_i$
3. Sum the magnitude of the weights corresponding to the sorted "signed" samples beginning with the maximum and continuing down in order.
4. The output is the signed sample whose magnitude weight causes the sum to become $\geq T_0$

2.2 Nonlinear Filtering for Image Analysis and Enhancement

- Weighted Median Filter: example

Observation samples	−2,	2,	−1,	3,	6
Corresponding weights	0.1,	0.2,	0.3,	−0.2,	0.1
Sorted signed observation samples	−3,	−2,	−1,	2,	6
Corresponding absolute weights	0.2,	0.1,	0.3,	0.2,	0.1
Partial weight sums	0.9,	0.7,	<u>0.6,</u>	0.3,	0.1

2.2 Nonlinear Filtering for Image Analysis and Enhancement

Weighted Median Filters for Color Images - Vector weighted median filter

- Each pixel includes 3 colors $\mathbf{x}_i = [x_i^1, x_i^2, x_i^3]^T$
- The weighted median of a set of 3-dimensional samples

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^N |W_i| \cdot || \text{sgn}(W_i) \mathbf{x}_i - \boldsymbol{\beta} ||$$

2.2 Nonlinear Filtering for Image Analysis and Enhancement - Applications

Image Zooming

- Applications
 - World Wide Web: for better visualization of images and content
 - Digital Video and DVDs: enhances viewing experience by resizing video frames without losing quality.
 - Scientific Imaging: Critical for detailed analysis by enlarging images for closer inspection.
- Process
 - New pixels are added to expand the size of the image

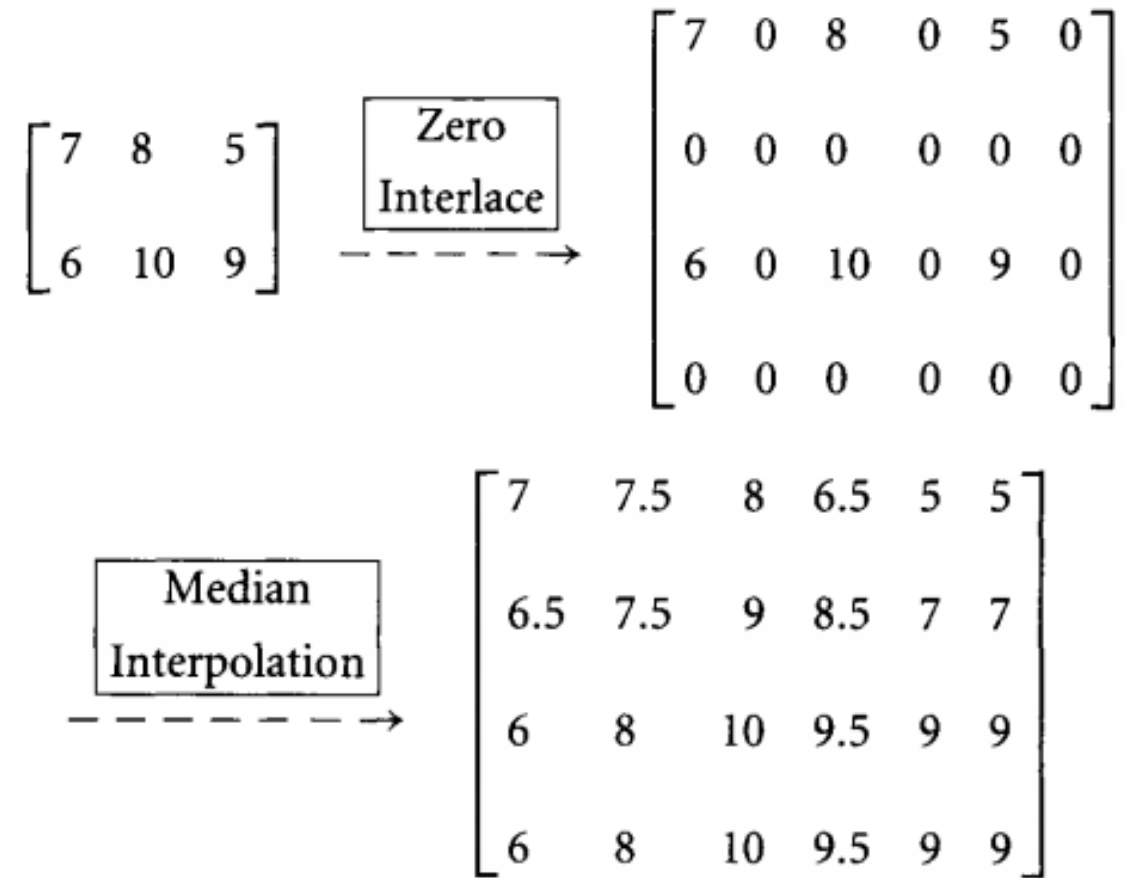
2.2 Nonlinear Filtering for Image Analysis and Enhancement - Applications

Image Zooming: Weighted Medians in Interpolation

- Interpolates new pixels based on the weighted median of surrounding pixels.
- Advantages over traditional linear methods
 - Edge preservation: Provides better edge-preservation, maintaining the sharpness of image edges
 - Reduction of artifacts: Minimizes the "blocky" appearance of edges often seen in linear interpolation methods.

2.2 Nonlinear Filtering for Image Analysis and Enhancement - Applications

- Image Zooming: zoomed by a factor of 2, and the median of the closest two (or four) original pixels is used to interpolate each new pixel



2.2 Nonlinear Filtering for Image Analysis and Enhancement - Applications

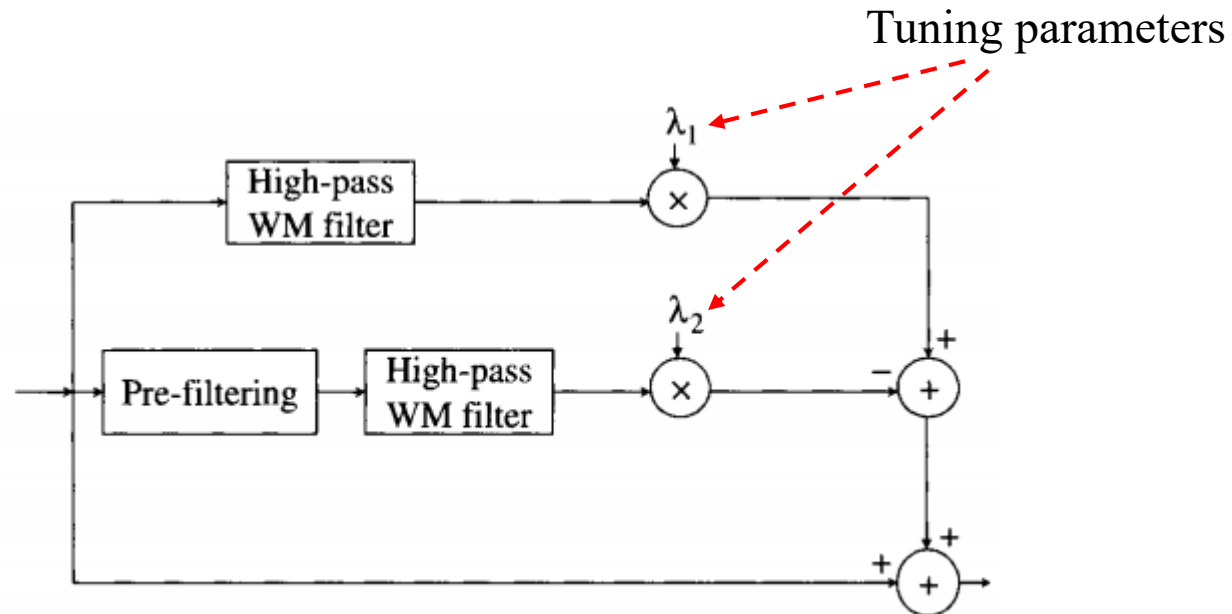
Image Sharpening

- Gray-scale image is filtered using the following a weighted median filter mask

$$W = \frac{1}{3} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

2.2 Nonlinear Filtering for Image Analysis and Enhancement - Applications

- Image sharpening based on the weighted median filter



λ_1 and λ_2 tuning parameters that control the amount of sharpness
generally selected to be equal

2.3 Morphological Filtering or Image Enhancement and Feature Detection

Mathematical morphology

- A nonlinear, set- and lattice-theoretic methodology for image analysis
- Aims to quantitatively describe the geometrical structure of image objects
- Overcomes the limitations of traditional linear methods, especially for problems involving geometric aspects of images
- Core operations are based on set theory, such as union, intersection, and complement.
- Utilizes lattice structures to analyze the relationships and hierarchies between different image elements.

2.3 Morphological Filtering or Image Enhancement and Feature Detection

Algorithms

- **Erosion:** Shrinks objects by removing pixels on object boundaries.
- **Dilation:** Expands objects by adding pixels to object boundaries.
- **Opening:** Erosion followed by dilation, useful for removing small noise.
- **Closing:** Dilation followed by erosion, useful for closing small holes and gaps.

2.3 Morphological Filtering or Image Enhancement and Feature Detection

Erosion algorithm

- Input:
 - Binary image $f(x,y)$ and Structuring element (SE) of size $m \times n$
- Initialize:
 - Create an output image $g(x,y)$ of the same size as the input image, initialized to 0.
- For each pixel (x, y) in the input image:
 - Center the structuring element SE at pixel (x,y) .
 - If **all** pixels under the SE in the input image are 1 (i.e., $f(x+i, y+j)=1$ for all (i,j) in the SE), set the corresponding pixel in the output image $g(x,y)$ to 1.
 - Otherwise, set $g(x,y)$ to 0.

2.3 Morphological Filtering or Image Enhancement and Feature Detection

Dilation Algorithm

- Input:
 - Binary image $f(x,y)$ and Structuring element (SE) of size $m \times n$
- Initialize:
 - Create an output image $g(x,y)$ of the same size as the input image, initialized to 0.
- For each pixel (x, y) in the input image:
 - Center the structuring element SE at pixel (x,y) .
 - If **any** pixels under the SE in the input image are 1 (i.e., $f(x+i, y+j) = 1$ for all (i,j) in the SE), set the corresponding pixel in the output image $g(x,y)$ to 1.
 - Otherwise, set $g(x,y)$ to 0.

2.3 Morphological Filtering or Image Enhancement and Feature Detection

Opening Algorithm

- Input:
 - Binary image $f(x,y)$ and Structuring element (SE)
- Perform Erosion:
 - Apply the erosion algorithm to the input image $f(x,y)$ using the structuring element SE.
 - Store the result in an intermediate image $f'(x,y)$
- Perform Dilation:
 - Apply the dilation algorithm to the eroded image $f'(x,y)$ using the same structuring element SE.

2.3 Morphological Filtering or Image Enhancement and Feature Detection

Closing Algorithm

- Input:
 - Binary image $f(x,y)$ and Structuring element (SE)
- Perform Dilation:
 - Apply the dilation algorithm to the input image $f(x,y)$ using the structuring element SE.
 - Store the result in an intermediate image $f'(x,y)$
- Perform Erosion:
 - Apply the erosion algorithm to the eroded image $f'(x,y)$ using the same structuring element SE.

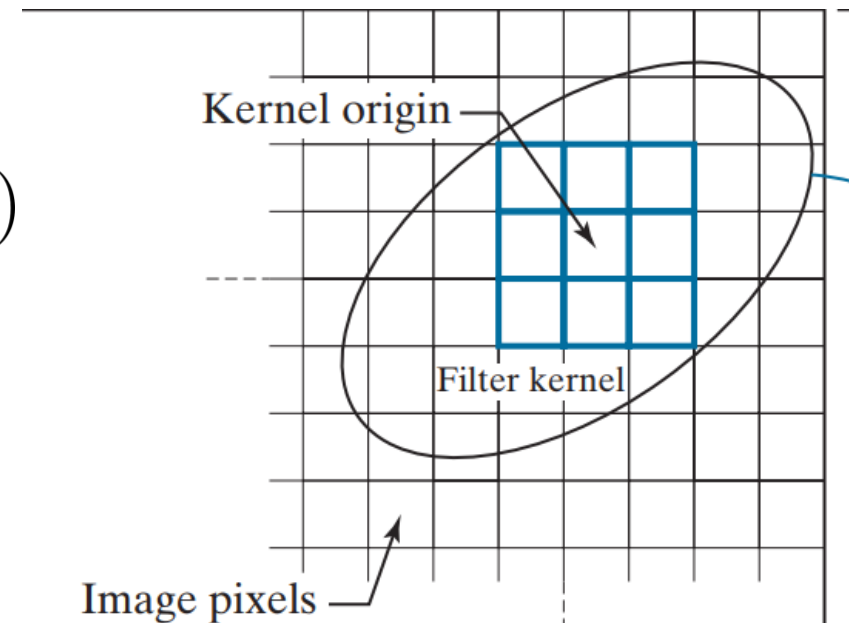
2.4 Spatial filters

- Spatial filtering modifies an image by replacing the value of each pixel by a function of the values of the pixel and its neighbors.
- If the operation performed on the image pixels is linear, then the filter is called a *linear spatial filter*.
- Otherwise, the filter is a *nonlinear spatial filter*.

2.4 Spatial filters – Linear spatial filtering

- A linear spatial filter performs a sum-of-products operation between an image f and a filter kernel, w
- The kernel is an array whose size defines the neighborhood of operation, and whose coefficients determine the nature of the filter.

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$



2.4 Spatial filters – Linear spatial filtering

- Spatial correlation consists of moving the center of a kernel over an image, and computing the sum of products at each location.
- The mechanics of spatial convolution are the same, except that the correlation kernel is rotated by 180° .
- When the values of a kernel are symmetric about its center, correlation and convolution yield the same result.

- Spatial correlation

$$(w \star f) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

- Spatial convolution

$$(w \star f) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x - s, y - t)$$

2.4 Spatial filters – Linear spatial filtering

- Properties of convolution and correlation

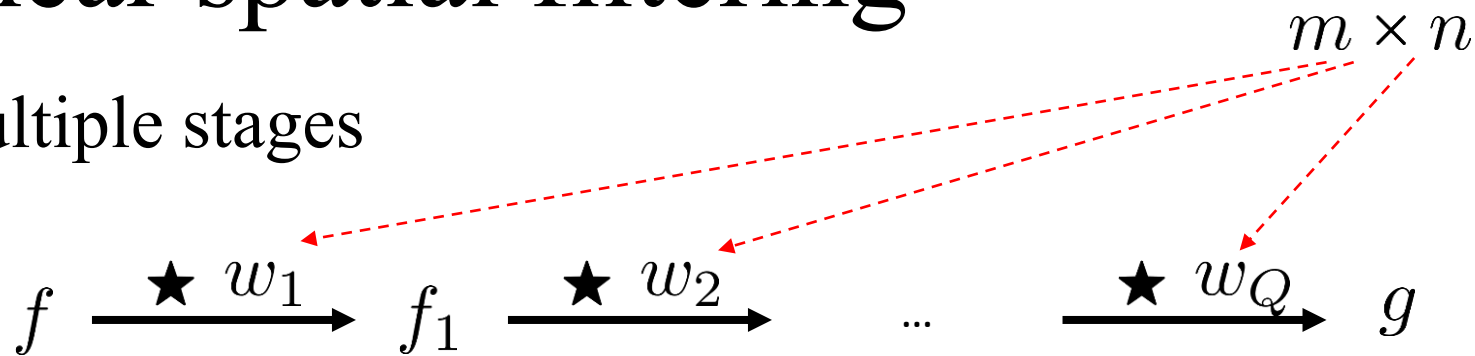
Property	Convolution	Correlation
Commutative	$f \star g = g \star f$	—
Associative	$f \star (g \star h) = (f \star g) \star h$	—
Distributive	$f \star (g + h) = (f \star g) + (f \star h)$	$f \star (g + h) = (f \star g) + (f \star h)$

- Size of resulting image

$$\begin{cases} S_v = m + M - 1 \\ S_h = n + N - 1 \end{cases}$$

2.4 Spatial filters – Linear spatial filtering

- Multiple stages



$$w = w_1 \star w_2 \star w_3 \star \cdots \star w_Q$$

$$\Rightarrow g = w \star f$$

- Size of aggregate filter kernel

$$\begin{cases} W_v = Q \times (m - 1) + m \\ W_h = Q \times (n - 1) + n \end{cases}$$

2.4 Spatial filters – Smoothing spatial filters (lowpass)

Box filter

- The simplest, separable lowpass filter
- The average value of an area of constant intensity would equal that intensity in the filtered image
- Normalizing the kernel in this way prevents introducing a *bias* during filtering; that is, the sum of the pixels in the original and filtered images will be the same

 $\frac{1}{9} \times$

1	1	1
1	1	1
1	1	1

box filter 3x3

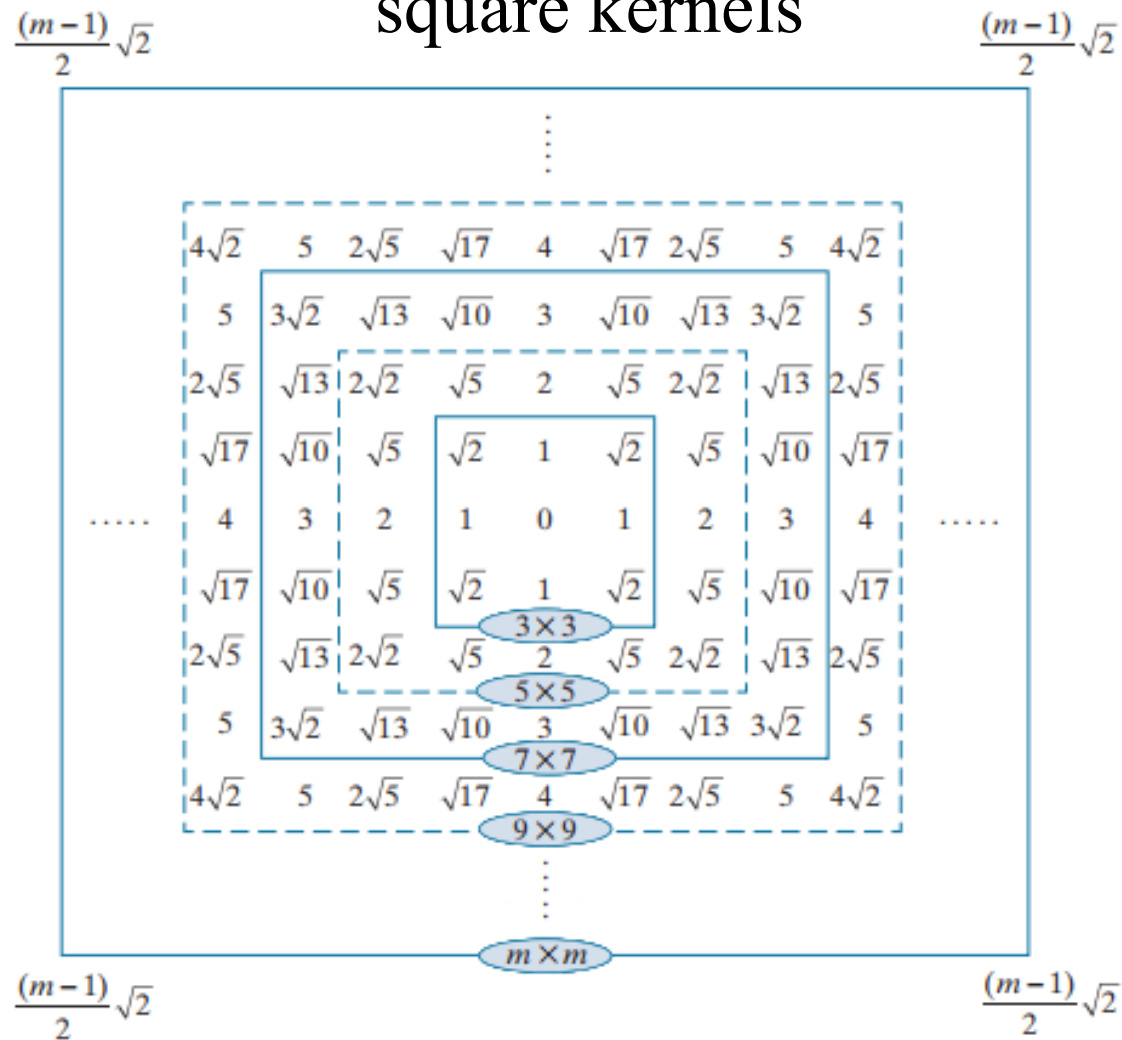
2.4 Spatial filters – Smoothing spatial filters (lowpass)

- Lowpass gaussian filter

$$G(r) = K e^{-\frac{r^2}{2\sigma^2}}$$

$$\frac{1}{4.8976} \times$$

0.3679	0.6065	0.3679
0.6065	1.0000	0.6065
0.3679	0.6065	0.3679

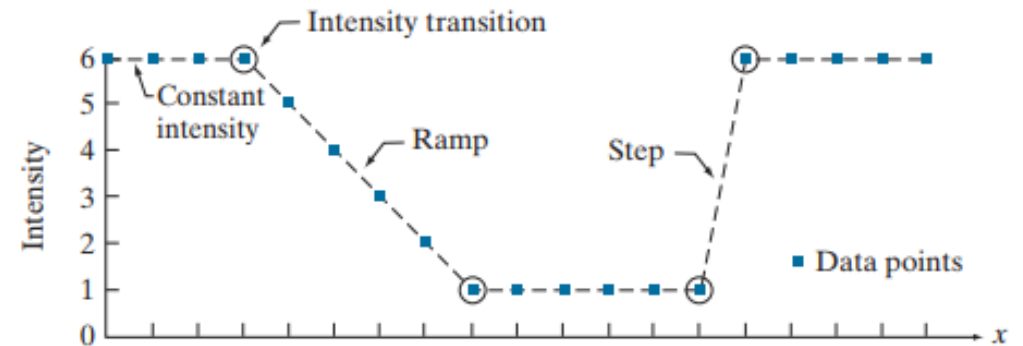


2.4 Spatial filters – Sharpening spatial filters (highpass)

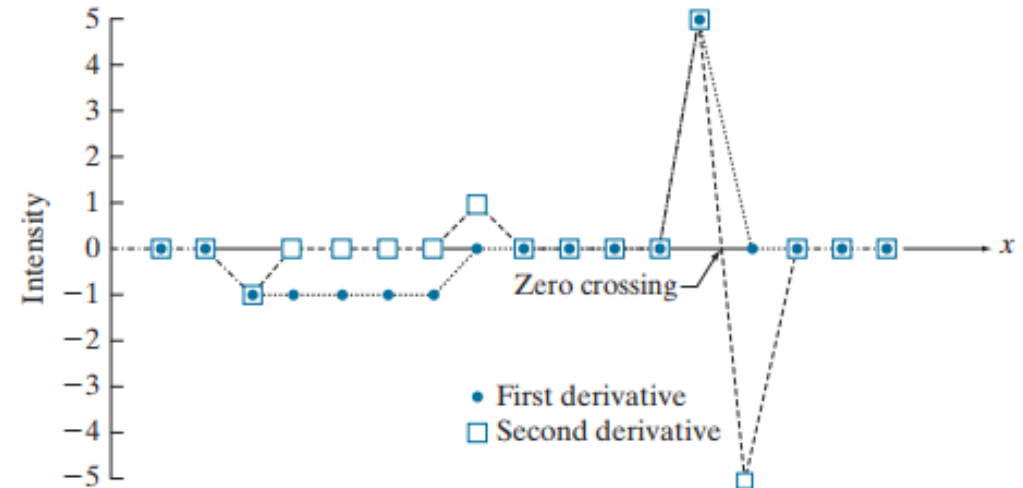
- In digital functions
- *first-order* derivative
 - Must be zero in areas of constant intensity.
 - Must be nonzero at the onset of an intensity step or ramp.
 - Must be nonzero along intensity ramps.
- *second-order* derivative
 - Must be zero in areas of constant intensity
 - Must be nonzero at the onset *and* end of an intensity step or ramp.
 - Must be zero along intensity ramps.

2.4 Spatial filters – Sharpening spatial filters (highpass)

- Visualization of definitions and example



Values of scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6	x
1st derivative	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0	0	
2nd derivative	0	0	-1	0	0	0	0	1	0	0	0	0	0	5	-5	0	0	0	0	



2.4 Spatial filters – Sharpening spatial filters (highpass)

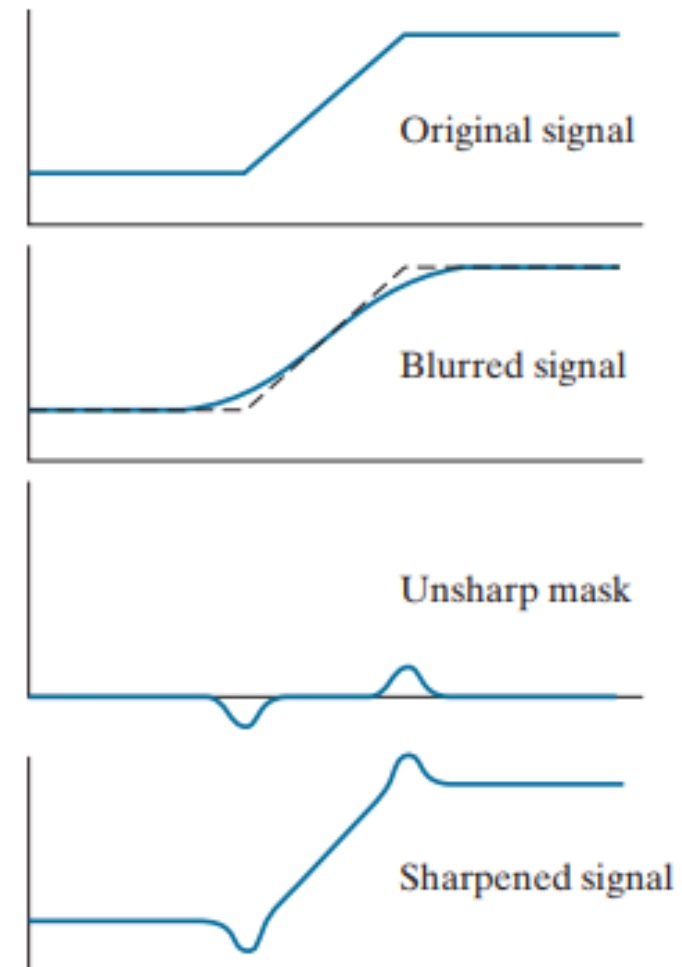
- Laplacian

$$\nabla^2 f(x, y) = f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1) - 4f(x, y)$$

0	1	0	1	1	1	0	-1	0	-1	-1	-1
1	-4	1	1	-8	1	-1	4	-1	-1	8	-1
0	1	0	1	1	1	0	-1	0	-1	-1	-1

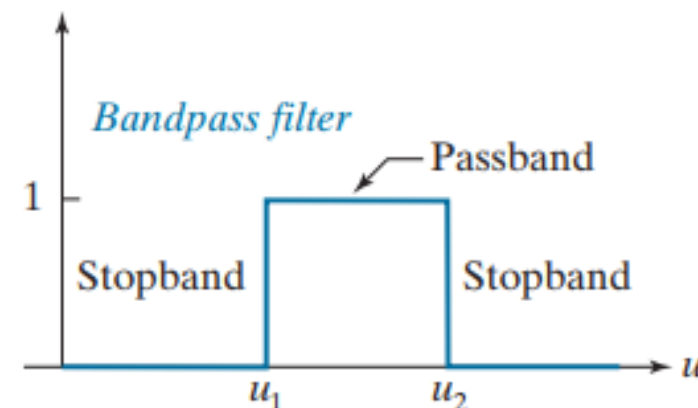
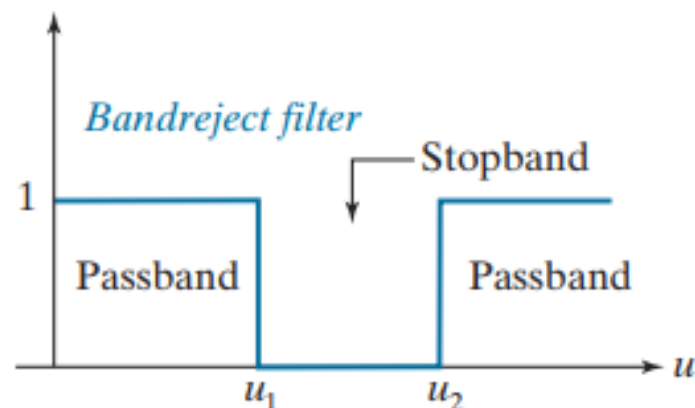
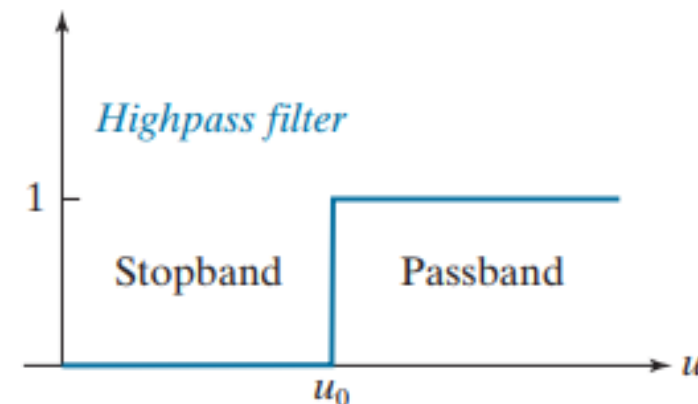
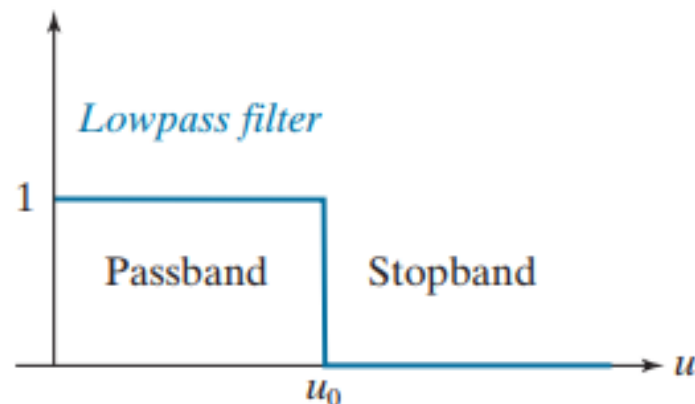
2.4 Spatial filters – Unsharp masking and highboost filtering

1. Blur the original image
2. Subtract the blurred image from the original (the resulting difference is called the *mask*.)
3. Add the mask to the original.



2.4 Spatial filters – Lowpass, Highpass, Bandreject, and Bandpass

- Express any filter from lowpass filter?



2.4 Spatial filters

- Discussion: Can spatial enhancement methods be combined to improve the result?

2.5 Filtering in the Frequency Domain – Discrete Fourier transform (DFT)

- 2-D discrete Fourier transform (DFT)

$$\mathfrak{F}\{f(x, y)\} = F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

- 2-D inverse discrete Fourier transform (IDFT)

$$\mathfrak{F}^{-1}\{F(u, v)\} = f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

2.5 Filtering in the Frequency Domain – Fourier spectrum and phase angle

Fourier spectrum and phase angle

- 2-D DFT is expressed in polar form:

$$F(u, v) = R(u, v) + jI(u, v) = |F(u, v)|e^{j\Phi(u, v)}$$

- *Fourier (or frequency) spectrum* $|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2}$
- *Phase angle or phase spectrum* $\Phi(u, v) = \arctan \left[\frac{I(u, v)}{R(u, v)} \right]$
- *Power spectrum* $P(u, v) = |F(u, v)|^2 = R^2(u, v) + I^2(u, v)$

2.5 Filtering in the Frequency Domain – Steps for filtering in the frequency domain

1. Given an input image $f(x,y)$ of size $M \times N$, obtain the padding sizes P and Q ; that is, $P = 2M$ and $Q = 2N$.
2. Form a padded image $f_p(x,y)$ of size $P \times Q$ using zero-, mirror-, or replicate padding
3. Multiply $f_p(x,y)$ by $(-1)^{x+y}$ to center the Fourier transform on the $P \times Q$ frequency rectangle.
4. Compute the DFT, $F(u,v)$, of the image from Step 3.
5. Construct a real, symmetric filter transfer function, $H(u,v)$, of size $P \times Q$ with center at $(P/2, Q/2)$.

2.5 Filtering in the Frequency Domain – Steps for filtering in the frequency domain

6. Form the product $G(u, v) = H(u, v)F(u, v)$ using element-wise multiplication; that is $G(i, k) = H(i, k)F(i, k)$ for $i = 0, 1, \dots, N-1$
7. Obtain the filtered image (of size $P \times Q$) by computing the IDFT of $G(u, v)$

$$g_p(x, y) = \left(\text{real} \left[\mathfrak{F}^{-1} \{ G(u, v) \} \right] \right) (-1)^{x+y}$$

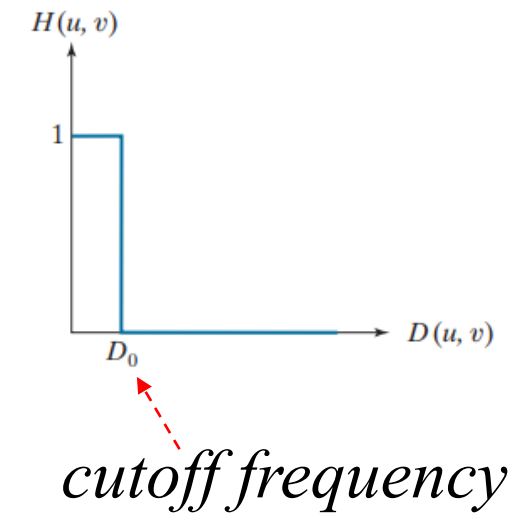
8. Obtain the final filtered result, $g(x, y)$, of the same size as the input image, by extracting the $M \times N$ region from the top, left quadrant of $g_p(x, y)$

2.5 Filtering in the Frequency Domain – Image smoothing (Lowpass filters)

- Ideal lowpass filter $H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$

- Gaussian lowpass filter $H(u, v) = e^{-\frac{D^2(u, v)}{2D_0^2}}$

- Butterworth lowpass filter $H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$



where $D(u, v) = [(u - P/2)^2 + (v - Q/2)^2]^{1/2}$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle, D_0 is a positive constant

2.5 Filtering in the Frequency Domain – Image sharpening (Highpass filters)

- Ideal highpass filter $H_{\text{HP}}(u, v) = 1 - H_{\text{LP}}(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$
- Gaussian highpass filter $H_{\text{HP}}(u, v) = 1 - H_{\text{LP}}(u, v) = 1 - e^{-\frac{D^2(u, v)}{2D_0^2}}$
- Butterworth highpass filter $H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$
- Laplacian $H(u, v) = -4\pi^2(u^2 + v^2)$

2.5 Filtering in the Frequency Domain – Applications



Smudged
thumbprint



Highpass
filtering



thresholding

2.5 Filtering in the Frequency Domain – Applications



Original blurry image



Laplacian based sharpening

Summary

- Basic Linear Filtering with Application to Image Enhancement
- Nonlinear Filtering for Image Analysis and Enhancement
- Morphological Filtering for Image Enhancement and Feature Detection
- Spatial Filtering
- Filtering in frequency domain